



Estd: 1986

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SJC INSTITUTE OF TECHNOLOGY

Chickballapur – 562 101

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Tutorial - 1

SUBJECT TITLE	Big Data Analytics		
SUBJECT TYPE	Integrated		
SUBJECT CODE	BAD601		
ACADEMIC YEAR		Batch	
SCHEME	NEP CBCS 2022		
SEMESTER	6th sem, AI&DS		
FACULTY NAME and DESIGNATION	Darshan A, Assistant Professor		
DATE of ISSUED			

Q. No.	Questions	Level	COs
1	Explain Structured data, semi structured data and unstructured data with examples. And also explain how to deal with unstructured data?	L2	CO1
2	Define Big Data and explain its importance. Describe the 3Vs of Big Data with detailed explanations	L2	CO1
3	Explain a typical data warehouse environment and a typical hadoop environment with a neat diagram.	L2	CO1
4	Explain big data analytics features and its importance. And also explain classification of analytics.	L2	CO1
5	Explain CAP theorem in detail.	L2	CO1
6	Explain below given terminologies used in big data environment: a) In-Memory analytics b) In-database processing c) Symmetric Multiprocessing (SMP) d) Massively Parallel Processor e) Shared Nothing Architecture f) Parallel system and distributed system	L2	CO1
7	Define NoSQL. List out and explain the features of NoSQL database. List out the applications of NoSQL.	L2	CO1
8	List out and explain different types of NoSQL databases.	L2	CO1
9	List out and explain the advantages of NoSQL.	L2	CO1
10	Differentiate the below given terms: a) SQL and NoSQL b) SQL, NoSQL and NewSQL c) HDES and HBase	L4	CO1

	d) Hive and RDBMS e) Hive and HBase f) Hadoop and SQL		
11	Define NewSQL. Explain the characteristics of NewSQL.	L2	CO1
12	List out and explain the features and advantages of Hadoop.	L2	CO1
13	List out and explain the version of Hadoop.	L2	CO1
14	Explain different stages that are involved in hadoop ecosystem.	L2	CO1
15	Explain the following: a) Sqoop b) Flume c) Mapreduce d) Spark e) Pig f) Hive	L2	CO1
16	Explain in detail about cloud based ecosystem.	L2	CO1
17	List out and explain the advantages of Hadoop. And also differentiate between Hadoop and RDBMS.	L4	CO2
18	Explain the challenges of distributed computing. And also explain how to process the gigantic store of data.	L2	CO2
19	Define Hadoop. List out and explain the functionalities of hadoop.	L2	CO2
20	List out and explain the key aspects of hadoop. And also explain about the components of hadoop in detail.	L2	CO2
21	Explain Hadoop conceptual layer and Hadoop High-level architecture with a neat block diagram.	L2	CO2
22	Explain the key components of master node in hadoop.	L2	CO2
23	List out and explain the features of HDFS. And also explain the HDFS architecture with a neat block diagram.	L2	CO2
24	Explain different types of HDFS daemons with neat diagram.	L2	CO2
25	Explain anatomy of file read with a neat block diagram.	L2	CO2
26	Explain anatomy of file write with a neat block diagram.	L2	CO2
27	Explain hadoop default replica placement strategy.	L2	CO2
28	Write the HDFS commands for the given tasks below: a) To get the list of directories and files at the root of hdfs b) To get the list of complete directories and files of hdfs. c) To create a directory in hdfs d) To copy a file from hdfs to a local file system. e) To copy a file from local system to hdfs via copyfromlocal command. f) To copy a file from local system to hdfs via copytolocal command. g) To display the contents of an hdfs on console. h) To copy a file from one directory to another on hdfs. i) To remove a directory from hdfs.	L3	CO2

Sign of the faculty
Darshan A

HOD